FoodJudgeBench: Benchmarking LLM Judges Against Human Annotators on Personalized Food Recommendation Explanations

1st Arshman Tariq (Team Leader)
Computer Science and Engineering
ITEE, University of Oulu
Oulu, Finland
2509523, atariq25@student.oulu.fi

2nd Faisal Khan
Computer Science and Engineering
ITEE, University of Oulu
Oulu, Finland
2512471, fkhan25@student.oulu.fi

3rd Muhammad Faizan Tanveer
Computer Science and Engineering
ITEE, University of Oulu
Oulu, Finland
2509524, mtanveer25@student.oulu.fi

***Abstract*—Large language models (LLMs) are now used both to write and to grade explanations for personalised food recommendations, but it is unclear whether their judgment can stand in for human annotation. We present FoodJudgeBench, a benchmark built on the HUMMUS recipe dataset that compares LLM judges with human annotators. We constructed 30 profiles inferred from food review histories and 30 fully specified synthetic profiles, selected two recipes per profile with a content-based recommender, and generated explanations with four techniques: template-based, direct LLM, evidence-grounded LLM, and evidence-grounded LLM with claim-level self-check and revision. The case bank contains 480 explanation cases plus 15 diagnostic cases with injected errors and 15 pairwise comparisons, graded with a ten-question rubric. Three open-weight judges from three model families (Gemma 2, Qwen 2.5, Phi-4) were run under several prompt setups. Specifications achieved: profile construction, recommendation pool, four explanation techniques, case bank, rubric, a 10-item pilot annotations with three annotators, LLM judgments, agreement analysis and a public release package. The full three-annotator main round on all 510 annotations is out of this project scope and will be done in the later stage. On the pilot, the best judge setups reached agreement with humans that is comparable to human-human agreement (within-one-point agreement 0.93 versus 0.88; yes/no kappa 0.35 versus 0.22), and judges ranked the four techniques similarly to humans. With only ten rated items the confidence intervals are wide, so these results demonstrate the benchmark pipeline and do not yet rank the judges.**

***Index Terms*—LLM-as-a-judge, explainable recommendation, food recommendation, HUMMUS, inter-annotator agreement, benchmark**

Project ID: 22.
Repository: https://github.com/arshmant25/FoodJudgeBench

I. Introduction

Food recommender systems help people choose what to cook or eat, and they increasingly have to respect health goals and practical limits such as time, diet and allergies [1], [2]. Explanations are a central part of such systems because they let users understand, trust and act on a recommendation [3], [4]. When the explanation also frames the choice, for example by emphasizing a health benefit, it acts as a nudge [5], which makes honesty and proportionality important requirements, not only fluency.

Large language models can now write such explanations on demand, and they are also used to evaluate them. The "LLM-as-a-judge" approach replaces or complements human raters with a model that grades outputs against a rubric [6], [7]. Work such as G-Eval [8] and Prometheus [9] shows that rubric-conditioned judges can correlate with human judgements on general text quality. At the same time, judges are known to have systematic weaknesses: they are sensitive to the order in which answers are presented [10] and they tend to favor text produced by their own model family [11]. These findings come mostly from open-domain tasks. Personalized recommendation explanations differ in an important way: a good grade depends on checking claims against a specific user profile and specific recipe facts, such as a time limit, a diet requirement, a nutrition number or a preference that the profile does or does not contain. A fluent explanation can invent a preference or misstate a nutrition fact without sounding wrong.

Problem and novelty. We ask whether LLM judges can grade food recommendations and their explanations as reliably as human annotators. We could not find a benchmark that (i) compares judges with multiple human annotators on personalized food explanations, (ii) separates profiles inferred from real histories from fully specified synthetic profiles, (iii) includes controlled injected errors to test whether judges detect invented personalization, factual errors and ignored constraints, and (iv) lets us measure the effect of giving the judge the checked facts. FoodJudgeBench addresses these points on top of the HUMMUS dataset [12], which links Food.com recipes and reviews [13] with nutrition and healthiness scores. The main contributions are:

- a reproducible pipeline from raw HUMMUS data to a blinded evaluation case bank with a hidden answer key;
- an evidence-grounded generation technique with claim-level self-check and revision, compared with template-based and direct LLM explanations;
- a ten-question rubric with explicit "not applicable" handling, offline annotation forms and agreement analysis (Krippendorff's alpha, Cohen's kappa);

- a judge harness with three prompt setups and three open-weight judge families, and an analysis notebook that compares judges with the human baseline using bootstrap confidence intervals;
- an honest pilot-scale evaluation of the four research questions in Section II.

II. Problem Description and Dataset

A. Problem description

The project specification defines four research questions. **RQ1:** how closely do LLM judgments agree with human assessments? **RQ2:** does agreement differ between profiles reconstructed from HUMMUS and synthetic profiles? **RQ3:** do LLM judges rank explanation techniques the way humans do? **RQ4:** which judging setup best detects unsupported personalization, factual errors and ignored constraints? The project evaluates assessment reliability only. There is no live experiment with the original users, and persuasion or behavior change is out of scope. Humans judge suitability for the supplied profile, not according to their own taste, and for profiles inferred from history they judge consistency with the available evidence and not the unknown person's true preference.

B. Dataset

We use the HUMMUS release (`pp_recipes`, `pp_reviews`, `pp_members`), obtained from the authors' repository [12]. HUMMUS extends the public Food.com interaction data [13] with linked ingredient information and three healthiness scores (WHO, FSA and a Nutri-Score style score). Its interactions are real user behavior, not ground-truth labels of explanation quality. Table I summarizes the files after our audit.

TABLE I
HUMMUS files and key audit results (Notebook 1).

File / item	Count	Note
Recipes	507,335	36 columns
Reviews (raw rows)	1,916,424	includes authorship rows
Members	299,583	87–90% missing profile fields
True reviews (rating 1–5)	1,409,914	used for all analysis
Authorship rows (rating 6)	506,510	author = recipe owner in 100%
Users with at least 10 true reviews	15,210	5.5% of reviewers
Rating 5 share	72.2%	mean 4.54, SD 0.86
Rating 2.5 rows	76,876	likely "no stars" placeholder

The audit changed how the data are used. First, the value 6 in the rating column does not encode a rating: all 506,510 such rows have no review identifier and the member equals the recipe author, so they are authorship records. We split them into a separate table, so that no later step counts them as interactions (Fig. 1). Second, the rating distribution is strongly skewed to 5 (72.2%), and 30.4% of users with at least ten reviews average 4.8 or more (Fig. 2). Absolute ratings are therefore a weak preference signal, so we use the rating relative to a user's own behaviour, a liked flag (rating 4 or above) and review text. Third, the value 2.5 appears with text and review identifiers like normal reviews, so we treat it as "unrated" and not as a mild dislike. Fourth, user histories are extremely sparse: the median user has one review, the 99th percentile has 62, and the user-recipe matrix is 99.998% empty. Given that absolute ratings are a weak preference signal. So, we use rating relative to the user's own mean (Fig. 3). Finally, the only timestamp is the last modification date of a review, not its submission date (Fig. 4), which limits any temporal hold-out; with a 2018 cutoff, only 316 users keep at least 15 earlier and 3 later reviews.

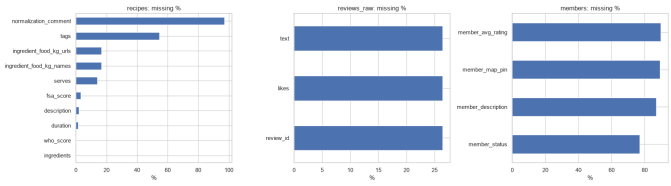


Fig. 1. Missingness in the three HUMMUS tables. `normalization_comment` (97.2%) and `tags` (54.5%) are the sparsest recipe fields, whereas the FSA score (3.1%) and WHO score (0.2%) are almost complete. Member profile fields are 77–90% missing, so they were not used.



Fig. 2. Number of reviews per rating value (true reviews only).

Most of the users gave 5 rating to the recipes. While, the 2.5 rating does not indicate the mild dislikes but that the users have written a review but did not rate the recipe.

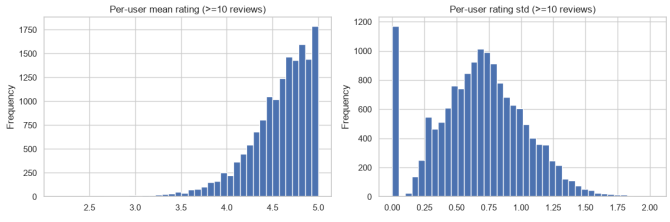


Fig. 3. Frequency of the number of true reviews per user.

276,377 users review a single recipe; a small group of heavy users has thousands of reviews and was excluded from the profile window (567 true users with more than 300 reviews).

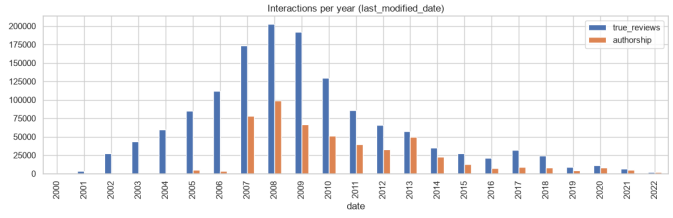


Fig. 4. True reviews and authorship records per year of last modification. Review activity peaks in 2008 (about 203,000 true reviews) and declines afterwards.

On the recipe side, we parsed the ingredient strings (stringified Python objects, not JSON), checked nutrition plausibility and the direction of the health scores, and applied quality filters (at least three ingredients and two steps, duration at most one day, at most 5,000 kcal per serving). The nutrition distributions are heavy-tailed (Fig. 5), so raw values are converted to percentiles in the catalog. The correlation between nutrients and the recipe attributes that matter for the benchmark is shown in Fig. 6, and the distribution of the healthiness scores in Fig. 7 and Fig. 8.

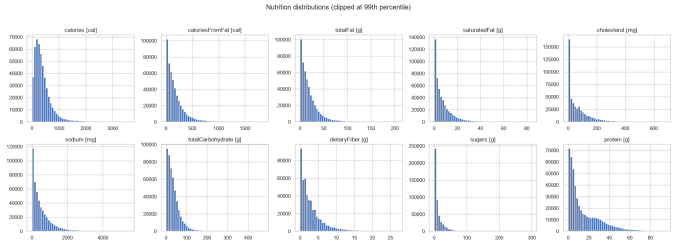


Fig. 5. Distribution of per-serving nutrition values in HUMMUS recipes (heavy-tailed, which motivates percentile ranks).

III. Methodology

A. General methodology

Fig. 9 shows the pipeline. Each stage is roughly one Jupyter notebook that reads the previous stage's files from `processed/` and writes its own, so every stage can be rerun and inspected independently. Notebook 1 audits and cleans HUMMUS; Notebook 2 builds the recipe catalog; Notebook 3 builds the two profile collections; Notebook 4 builds the recommendation pool; Notebook 5 generates explanations; Notebook 6 builds the blinded case bank, rubric and annotation forms; Notebook 8 collects and analyses human annotations; Notebook 9 runs the LLM judges; Notebook 10 compares judges with humans and writes the release benchmarks.
Table II maps each specification of the project plan to its notebook and report section.

B. Recipe catalog (Notebook 2)

Recipes that pass the quality filters (448,652) are reduced to 102,317 candidates with enough reviews and then to a stratified catalog of 6,522 recipes, sampled so that healthy and unhealthy recipes and short and long recipes are all present

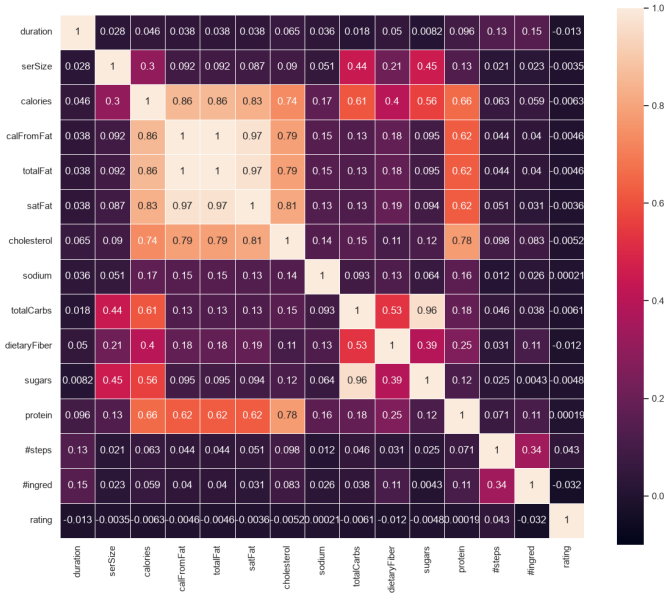


Fig. 6. Correlation matrix of nutrients and the recipe columns used by the pipeline. Nutrients are highly correlated to each other.

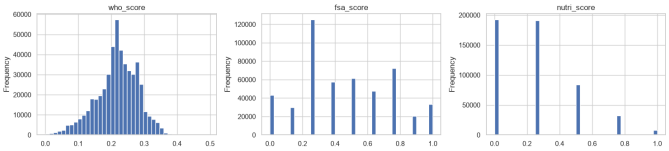


Fig. 7. Frequency of the healthiness score levels.

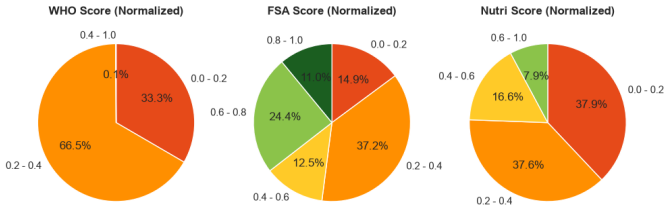


Fig. 8. Distribution of the WHO and FSA healthiness scores across recipes. 99.9% of the WHO scores lie between 0.0 and 0.4.

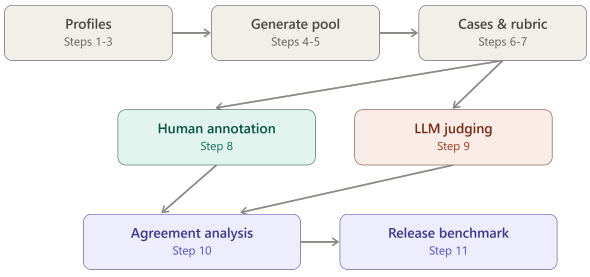


Fig. 9. FoodJudgeBench pipeline from the HUMMUS files to the released benchmark. Hidden identifiers (technique, model, error type) are kept in an answer key that annotators and judges never see.

TABLE II
PROJECT SPECIFICATIONS, IMPLEMENTING NOTEBOOKS AND REPORT SECTIONS.

Specification	Notebook (output)	Section
Step 1, 2: data audit and catalogs	1, 2 (`reviews_clean`, `catalog`)	II-B, III-B
Step 3: profiling	3 (`profiles_*.jsonl`)	III-C
Step 4: recommendation pool	4 (`recommendation_pool.jsonl`)	III-D
Step 5: explanation techniques	5 (`explanations.jsonl`)	III-E
Step 6: evaluation cases	6 (`case_bank.jsonl`)	III-F
Step 7: human rubric	6 (`rubric.json`)	III-G
Step 8: annotation analysis	8 (`annotations_*.csv`)	III-H
Step 9: LLM judges	9 (`judgments_*.csv`)	III-I
Step 10: agreement analysis	10 (`results/*.csv`)	III-J, IV
Step 11: release	10 (`release/`)	III-K

(minimum recipes per health level and duration band). Nutrition and duration are converted to percentile ranks, so the text can say "high in sodium compared with other recipes" with a defined meaning. We build review-weighted baselines: users review popular recipes, so testing a user's tendencies against the whole catalog would inflate z-scores. Diet constraints (vegetarian, vegan, dairy-free, gluten-free, nut-free) are labeled from ingredient rules and tags, with three states: *violates* (rule fails), *verified* (rule passes and a tag confirms it) and *rule_only*. The rules agree with the tags in 95.8% of vegetarian recipes (n = 35,599), 86.2% of vegan, 84.0% of gluten-free and 85.1% of dairy-free recipes, so they are approximate, which is why the labels of all selected recipes were hand-checked (Section III-D). The detailed stratum categorised by nutritional score and recipe duration can be reviewed in the table below. Three non-food entries (play dough, finger paint) were removed later step but are included in Table III.

TABLE III
STRATA OF CATALOG AS REPORTED IN `CATALOG_REPORT.JSON`.

Nutri-Score	Recipe Duration (Minutes)			Total
	$\leq$ 30	31–60	$>$ 60	
0	711	787	727	2,225
1	994	749	619	2,362
2	515	293	200	1,008
3	180	150	150	480
4	150	150	150	450
Total	**2,550**	**2,129**	**1,846**	**6,525**

C. Profile collections (Notebook 3)

All profiles share one schema in which every attribute carries a status (stated, inferred or unknown) and a link to its evidence.

Historical profiles. Thirty users were chosen from the window of 20–300 true reviews (7,673 users). A user needs rating variation (SD of at least 0.5), at least 10 liked recipes, at least one year of history and at least 15 text reviews. Users are stratified by history richness (sparse: 20–50 reviews; rich: 100–300 reviews) and by whether the reviews contain evidence of restrictions or health talk, giving 8, 8, 7 and 7 users per stratum. Likes and dislikes are ingredients and tags that are over-represented among the user's liked (or disliked) recipes compared with the review-weighted baseline (z-test and minimum lift); nutrition tendencies ("tends to like higher fiber") need at least ten recipes and a z-score above 3; a diet pattern is only reported if at least 95% of liked recipes satisfy it. No profile states a confirmed constraint: everything is marked as inferred, and the health goal stays unknown unless the history clearly supports a hint (three profiles show a "possibly lighter eating" hint). `profiles_historical.json` snapshot can be reviewed in Table IV.

TABLE IV
HISTORICAL PROFILE CHARACTERISTICS AND INTERACTION METRICS.

ID	Label	N_{reviews}	N_{liked}	Coverage	Top Likes	Tendencies
H001	rich/ no_signal	173	164	0.942	cookies, desserts	High sat. fat; low fiber
H002	rich/ no_signal	100	96	0.930	appetizers, mushrooms	High cooking time
H003	rich/ no_signal	292	271	0.904	brie, curries, tomatoes	Low cooking time

Synthetic profiles. Thirty personas are specified explicitly in four families: 14 standard (for example busy vegetarian, sodium watcher, nut-free family), 8 conflict (for example low sodium but loves salty food), 6 missing-information (for example a goal but no taste, or only dislikes) and 2 explicit "no restrictions" baselines (see Table V). Each persona is checked for feasibility, i.e. how many catalog recipes satisfy its hard constraints, its likes and its goal; thin cases are flagged (for example S016 has only two recipes meeting all three conditions). Explicit "unknown" fields are kept, so that the explanation must not assume them.

TABLE V
SYNTHETIC PROFILE METRICS.

ID	Label	$N_{\text{hard_feasible}}$	N_{verified}	$N_{\text{hard_liked}}$	$N_{\text{hard_goal}}$	N_{all3}
S002	standard/ sodium_watcher	3,812	3,812	941	1,589	235
S016	conflict/quick_ vegetarian_protein	331	59	33	11	2
S024	missing_info/ taste_only_spicy	6,525	6,525	1,766	–	–

D. Recommendation pool (Notebook 4)

For each of the 60 profiles the notebook builds a pool of about five candidates (300 in total) with a content-based method: recipes are described with TF-IDF vectors [14]; for historical profiles the query is the centroid of liked recipes minus a weighted centroid of disliked ones, and for synthetic profiles the score combines goal match, liked terms and recipe

quality with weights that depend on what the persona specifies. Hard constraints (diet and time) filter candidates first. Each candidate has a *role*: three *strong* matches (with a maximal-marginal-relevance rule so that they are not near duplicates), one *weak_taste* candidate, one *weak_goal* candidate and, for synthetic profiles, planned *violates_constraint* candidates that are used only as diagnostics. The pool therefore contains honest weak matches, because explanations of weak matches are where trade-offs have to be communicated.

Two candidates per profile become evaluation cases through the flags `selected_for_case` and `case_slot`: slot 1 is always a strong match; slot 2 cycles through weak_goal, weak_taste and a second strong match, with fallbacks (See Table VI for details). The result is 120 cases over 60 profiles (80 strong, 20 weak_goal and 20 weak_taste recipes). Candidates whose diet label is ambiguous or violated are replaced by the next strong candidate (eight replacements in the final run), after 50 manual label overrides from a hand-check sheet (21.7% of the selected labels were rule-only before checking).

TABLE VI
SELECTION FROM THE POOL OF PROFILES.

Profile	Recipe	Role	Rank	Score	Selected	Slot	Key Facts
H001	302333	strong	1	0.997	True	1.0	matches liked: desserts, peanut butter
H001	139560	strong	2	0.995	False	–	matches liked: cookies, brownies
H001	392340	strong	3	0.992	False	–	matches liked: cookies, brownies
H001	409321	w_taste	1	0.183	False	–	no matches, no dislikes
H001	440721	w_goal	1	0.971	True	2.0	matches liked: desserts

E. Explanation techniques (Notebook 5)

Every case gets four explanations (Table VII). Each case also has a numbered fact list (recipe facts, profile facts with status, constraint checks, time check, goal check and nutrition percentiles). Facts that describe a mismatch (for example an unmet goal) are marked *must mention*.

TABLE VII
THE FOUR IMPLEMENTED EXPLANATION TECHNIQUES.

Technique	How the text is produced
Template-based	Verified facts are inserted into fixed sentence patterns; no language model.
Direct LLM	The model sees the profile text, the recipe, its ingredients and nutrition, and writes a free explanation.
Grounded LLM	The model may use only the numbered facts, must cite them, state *must mention* items as trade-offs and phrase inferred facts cautiously.
Grounded + self-check	As above, then an auditor extracts each claim, labels it supported, unsupported or contradicted and lists missing trade-offs; the text is revised (at most twice) until no issue is left.

Text is generated by Llama 3.1 8B [15] through a local Ollama server at temperature 0.4; the self-check auditor is Qwen 2.5 14B [16]. The auditor is a different model from the generator, so that the generator does not check its own work. Fig. 10 summarizes the check-and-revise loop.

$T_0 \leftarrow$ GENERATEGROUNDED(facts)
for $r = 0, 1, 2$ **do**
  $C \leftarrow$ AUDIT(T_r, facts) {claims with verdicts}
  if no claim is unsupported or contradicted and no must-mention fact is missing **then**
    return T_r
  end if
  $T_{r+1} \leftarrow$ REVISE(T_r, C, facts)
end for
return T_2 {flag as unresolved}

Fig. 10. Pseudocode of the self-check grounded technique (technique 4).

The details of LLM explanations using the four mentioned LLM techniques are given as an example in (Table VIII) for only the first historical profile, first recipe.

TABLE VIII
LLM EXPLANATIONS FOR PROFILE H001, RECIPE 1.

Profile	Recipe	LLM Technique	Explanation
H001	302333	**template**	Sweetslady's Chocolate Peanut Butter Brownie Trifle ... 8 g of protein and 272 mg of sodium.
H001	302333	**direct_llm**	I think you'll love Sweetslady's Chocolate Peanut ... powdered sugar aligns with your preferences.
H001	302333	**grounded_llm**	Based on your past reviews, tend to like recipes that feature desserts, ... longer than your typical recipe.
H001	302333	**grounded_check_llm**	Based on your past reviews, tend to like recipes that feature desserts, ... longer than your typical recipe.

F. Evaluation cases and diagnostics (Notebook 6)

Notebook 6 joins profiles, recipe facts and the four explanations into a `case_bank.jsonl` file. The 480 normal cases (120 recipe-profile pairs times four techniques) are split evenly over historical and synthetic profiles (240 each). Each item gets a random identifier; technique, model, profile source and error type are removed from everything an annotator or judge can see and kept in a hidden answer key. A check confirms that no technique or model name remains in visible text. Cosmetic repairs only touch formatting artefacts of the generators (for example fractions lost in ingredient lines, a broken sentence start, or fact identifiers such as "[F3]" leaking into text); 98 of 480 texts and 39 profile texts were changed this way. We did not repair factual or personalisation errors: they remain in the text because they are what the benchmark measures.

Besides these 480 cases, we added fifteen *diagnostic* cases containing a controlled error injected into an otherwise normal explanation: an invented preference (4), a wrong nutrition comparison (4), an ignored time limit (4), and an ignored dietary constraint (3, built from recipes whose diet label is certainly violated after hand-checking). Also, 15 pairwise cases were added to compare two techniques for the same profile and recipe. This makes a total of 510 evaluation cases.

G. Human rubric (Notebook 6)

Humans rate suitability for the supplied profile, not their own taste. The rubric (Appendix Table XVIII) has five violation questions (yes/no, with "not applicable" where the question can be void) and five 1–5 scales. The violation questions are ignored constraint, factual error, invented personalization, unsupported claim and overstated certainty (presenting something inferred as certain). The scales are clarity, profile fit, goal alignment (not applicable when the profile states no goal), trade-off honesty and overall quality. Each annotator also has a free comment field, and pairwise items use "A, B or tie" with a reason. An annotator guide explains when "not applicable" is allowed; for example, the constraint question is not applicable only if the profile states neither a diet or allergy nor a time limit. These details can be best understood by the snapshot of the annotator form in the Appendix (Fig. 14).

H. Human annotation (Step 7 and Notebook 8)

Each annotator receives an offline HTML form (see Fig. 14) that autosaves in the browser and exports a CSV file, with items in a different random order per annotator. The plan is three independent annotators per normal case (A1, A2, A3 each rate all 510 items). Following a supervisor decision, the full main round was deferred, and a 10-item pilot (8 normal cases and 2 diagnostics, no pairs) was rated by all three annotators. Notebook 8 validates the files, merges annotators, and reports (i) Krippendorff's alpha, nominal for yes/no and ordinal for scales, (ii) mean pairwise Cohen's kappa (quadratic-weighted for scales), (iii) exact agreement and the share of scores within one point, (iv) the yes-rate, (v) an audit of "not applicable" use against the profile facts, (vi) the most disputed items and (vii) the share of diagnostics that were flagged. Krippendorff's alpha [17] is

$$\alpha = 1 - \frac{D_o}{D_e}, \tag{1}$$

where D_o is the observed and D_e the expected disagreement under chance. Cohen's weighted kappa [18] is

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij}\, O_{ij}}{\sum_{i,j} w_{ij}\, E_{ij}}, \quad w_{ij} = \frac{(i-j)^2}{(K-1)^2}, \tag{2}$$

with observed and chance-expected rating matrices O and E over K levels. For yes/no questions, unweighted kappa is computed only on items where both raters answered yes or no, and agreement on "not applicable" is reported separately. The pooled kappa in an earlier version of the notebook mixed rater pairs and was discarded.

I. LLM judges (Notebook 9)

Judges see the same evidence and rubric as humans: the profile (with stated or inferred labels), the recipe, an optional block of checked facts and the explanation, and they return one JSON object with the ten answers. Three setups were designed. *rubric_facts* is the human-equivalent setup (profile, recipe, checked facts and rubric). *rubric_nofacts* is an ablation without the checked-fact block, so the judge must rely on the recipe and ingredient list. *checklist* first asks the judge to list the claims of the explanation and the profile requirements and then to answer the rubric. Pairwise cases are judged in both orders to measure position bias.

Three judge families were used, all different from the generator family so that no judge grades its own text: Gemma 2 9B [19], Qwen 2.5 14B [16] and Phi-4 [20], all served locally by Ollama. Qwen also acted as the auditor of technique 4, which we discuss as a limitation. The temperature is 0.1 with a fixed seed, the context length is 4,096 tokens, and an HTTP 500 error from the server (a repetition guard) triggers up to three retries with a higher temperature and a new seed. Answers that break the format are repaired in a documented way (for example "na" on a plain yes/no question becomes "no") and flagged in a `coerced` column; in the reported run no answer needed repair. Results are cached on disk and the run is resumable. A first version used Mistral 7B, which failed to produce parseable answers for 30–70% of the items and was replaced by Phi-4. Fig. 11 summarizes the loop.

```
I ← items rated by at least one human
for all judge j, setup s, item i ∈ I do
    p ← BUILDPROMPT(i, s)
    a ← CACHE(j, p) or QUERY(j, p) with retry
    validate and, if needed, repair a; log coerced
    append row (judge|setup, item, answers) to the table
end for
```

Fig. 11. Pseudocode of the judge harness. Judges only grade items that humans rated.

J. Agreement analysis (Notebook 10)

Each judge|setup is treated as one more annotator. For every judge and every human annotator, we compute agreement on the same items and average the result; the baseline is the agreement between pairs of human annotators on the same items. Yes/no questions are compared with kappa and F1 of the "yes" (problem found) label; scales with the share of scores within one point,

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\left[\, |a_i - b_i| \le 1 \,\right], \tag{3}$$

Spearman correlation, quadratic-weighted kappa and mean absolute error. Confidence intervals are 95% percentile intervals from 300 bootstrap resamples [21] of items. RQ2 repeats the analysis for historical and synthetic profiles separately; RQ3 compares the mean overall score of each technique and the Spearman correlation of the technique ranking with the mean human ranking; RQ4 reports how often each judge flags the injected error type; pairwise outputs measure preference agreement and position bias (no human pairs were rated in the pilot, so this part is untested).

K. Release (Notebook 10)

The release folder contains the case bank without recipe text by default (Food.com terms), the rubric, the human and judge ratings, the answer key, result tables and figures, and a README (`insights.md`). The `data/` folder and the `.env` file are never published.

IV. Results and Discussion

All results in this section come from the **pilot**: 10 items (8 normal cases and 2 diagnostics), each rated by three annotators and by every judge setup. They demonstrate that the pipeline works end to end, but they are too few to rank judges (bootstrap intervals are wide, and a few flipped answers change kappa a lot). Rating pairs from one item are not independent, so the sample is effectively 10, not 510.

A. Generation outputs

Table IX shows the generated text. Template explanations are shorter (57 words on average) and the three LLM techniques are around 80 words. Only 1.7% of direct LLM texts contained a number that is not in the supplied facts, compared with 0% for the grounded techniques. In the self-check technique the auditor raised issues in round 0 for 35% of the cases (43 unsupported claims, 12 contradicted claims and 4 missing trade-offs in total); after at most two revisions 8 unsupported claims and 1 contradicted claim remained in 7 cases. Inspection shows that some of these remaining items are auditor mistakes, i.e. claims that the text never makes. Because the revision often changes nothing, technique 4 is identical to technique 3 in 83 of 120 cases (69%), which reduces the contrast between the two grounded techniques. Direct LLM texts use first-person phrasing ("I think you'll love…") in 72% of cases against 16% for the grounded techniques and 0% for templates, so the developers performing annotations may recognise the technique despite blinding.

TABLE IX
Generated explanations (120 cases per technique).

Technique	Words (mean)	Range	Number not in facts	First person
Template	56.7	28–87	0.0%	0.00
Direct LLM	82.8	63–98	1.7%	0.72
Grounded LLM	82.6	57–113	0.0%	0.16
Grounded + check	78.9	47–115	0.0%	0.16

B. Human annotation quality

Table X shows agreement among the three annotators. Alpha is moderate for invented personalization (0.48), trade-off (0.67) and overall (0.61), and low for ignored constraint (0.10), unsupported claim (0.28) and clarity (0.29). Goal alignment has alpha 0.00 although exact agreement is 0.80, because 73% of the answers are "not applicable" and the few numeric answers vary little. The ignored-constraint yes-rate is only 5%, so alpha is unstable. The audit of "not applicable" found some misuse (for the constraint question, two answers of "not applicable" although the profile stated a requirement and five answers although none applied), which led to a clearer rule in the annotator guide. Six items were disputed on four to seven of the ten questions. The median time per item was 370 s, with large differences among annotators (110 s, 424 s and 705 s), and the three annotators calibrate differently: one is consistently stricter on profile fit and overall quality. We interpret this as normal variation in judgment scale rather than annotation error, but the differences motivate a calibration session before the main round.

TABLE X
Human agreement on the pilot (10 items, three annotators).

Question	Exact	Alpha	Kappa	Within 1
Ignored constraint	0.57	0.10	0.16	–
Factual error	0.73	0.39	0.37	–
Invented personalization	0.73	0.48	0.48	–
Unsupported claim	0.67	0.28	0.27	–
Overstated certainty	0.57	0.34	0.32	–
Clarity	0.33	0.29	0.31	0.87
Profile fit	0.23	0.48	0.59	0.80
Goal alignment	0.80	0.00	0.00	1.00
Trade-off	0.30	0.67	0.67	0.79
Overall	0.43	0.61	0.62	0.93

C. RQ1: judge-human agreement

Table XI compares each judge setup with humans on the same ten items in detail, together with the human-human baseline. The results should be read with the intervals in mind.

Four observations stand out. (i) Several judge setups are statistically indistinguishable from the human baseline: Qwen with the checklist has a yes/no kappa (0.35) and a within-one-point agreement (0.93) at or above the human baseline (0.22 and 0.88), and Phi-4 with facts reaches kappa 0.28. (ii) Removing fact-grounding degrades performance sharply across all models; when operating with "no facts," Gemma 2 9B, Phi-4, and Qwen 2.5 14B see their Spearman correlations drop to their lowest respective levels (0.13, 0.30, and 0.39), with Gemma and Phi-4 experiencing severe spikes in Mean Absolute Error (MAE to 1.08 and 1.23). (iii) Gemma has the lowest overall yes/no agreement (kappa 0.08 for both facts and no facts), primarily because it almost never flags a problem. (iv) The human baseline itself is low for yes/no questions (kappa 0.22 with an interval from -0.02 to 0.41), so a judge that "matches the human level" is matching a modest level; the questions with few "yes" answers (ignored constraint, overstated certainty) have unstable kappas, and for them exact agreement and counts are more informative.

Fig. 12 breaks the agreement down by question (kappa for yes/no questions, Spearman correlation for scales). Judges agree best with humans on invented personalisation (Phi-4 checklist 0.67, Qwen checklist 0.73, against 0.47 between humans), on trade-off (0.69–0.79, humans 0.68) and on overall quality (0.54–0.76, humans 0.60). They agree least on overstated certainty and ignored constraint. Blank cells are not missing results: they mean that the statistic is undefined because one side gave a constant answer (for example Phi-4 gave goal alignment 4 for both items that had a goal, and Gemma almost always answered "no" or "na" to the constraint question). Goal alignment has only six numeric answers per judge. The scores also show how the evaluation setups differ. Gemma with facts has a factual-error score of 0.00 and an unsupported-claim score of 0.18, compared with 0.37 and 0.25

TABLE XI
RQ1: AGREEMENT WITH HUMAN ANNOTATORS ON THE 10 PILOT ITEMS (95% BOOTSTRAP INTERVALS IN BRACKETS).

Rater	Yes/no exact	Yes/no kappa	Yes F1	Within 1	Spearman	QW kappa	MAE
Human vs human (baseline)	0.65	0.22 [-0.02, 0.41]	0.42	0.88 [0.78, 0.95]	0.36	0.38	0.78
Gemma 2 9B, facts	0.59	0.08 [0.00, 0.23]	0.14	0.85 [0.74, 0.91]	0.38	0.27	0.83
Gemma 2 9B, checklist	0.65	0.12 [-0.01, 0.26]	0.30	0.84 [0.70, 0.91]	0.39	0.30	0.83
Gemma 2 9B, no facts	0.59	0.08 [-0.04, 0.26]	0.18	0.76 [0.61, 0.85]	0.13	0.10	1.08
Phi-4, facts	0.69	0.28 [0.04, 0.55]	0.43	0.85 [0.79, 0.90]	0.54	0.36	0.83
Phi-4, checklist	0.64	0.16 [0.02, 0.37]	0.38	0.83 [0.75, 0.87]	0.56	0.33	0.86
Phi-4, no facts	0.62	0.14 [-0.01, 0.33]	0.37	0.66 [0.49, 0.83]	0.30	0.15	1.23
Qwen 2.5 14B, facts	0.62	0.20 [0.00, 0.39]	0.43	0.78 [0.61, 0.94]	0.44	0.46	0.84
Qwen 2.5 14B, checklist	0.67	0.35 [0.21, 0.49]	0.45	0.93 [0.88, 0.97]	0.68	0.53	0.60
Qwen 2.5 14B, no facts	0.59	0.21 [0.08, 0.34]	0.46	0.62 [0.51, 0.86]	0.39	0.25	1.29

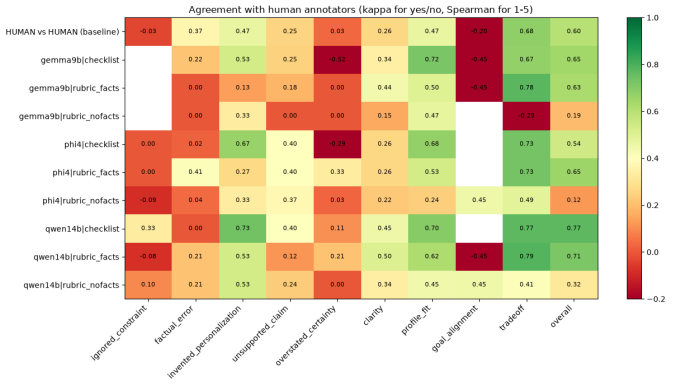


Fig. 12. Agreement of each judge setup with human annotators per rubric question (kappa for yes/no questions, Spearman for scales). Blank cells are undefined statistics.

for the human baseline. Its checklist variant increases these scores to 0.22 and 0.25, respectively, while its overstated-certainty score is -0.52. Phi-4 with facts scores 0.41 on factual errors and 0.40 on unsupported claims, whereas its checklist variant scores 0.02 and 0.40 on these dimensions. For Qwen, the checklist produces the highest invented-personalisation score (0.73) and an unsupported-claim score of 0.40. In comparison, Qwen with facts scores 0.12 on unsupported claims and 0.21 on overstated certainty. For ignored constraints, the human baseline is -0.03, while the model scores range from -0.09 to 0.33 where values are available.

Effect of the setup and cost. The checklist setup improves Qwen's scale agreement (within one point, from 0.78 to 0.93), but does not clearly improve agreement for Gemma or Phi-4. It also requires three to four times as much time: approx. 7 s (Gemma), 17 s (Qwen), and 20 s (Phi-4) per item for the rubric setup, compared with 23 s, 65 s, and 78 s, respectively, with the checklist on our hardware. For the full 510-item round, this corresponds to approximately 1 h versus 3 h for Gemma and 2.5 h versus 10 h for Qwen. Fig. 12 also indicates that removing the facts block does not produce a uniform change. These differences indicate that the effect of the facts block varies by model, evaluation dimension and run-to-run variation at this sample size.

D. RQ2: historical and synthetic profiles

Table XII compares agreement across historical (H) and synthetic (S) profiles. With six historical and four synthetic items, these results should be treated as indicative rather than conclusive. For humans, within-one-point agreement and Spearman correlation are higher on synthetic profiles (0.91 and 0.44) than on historical ones (0.83 and 0.24), as expected when preferences are stated and not inferred. A similar pattern appears for several model setups. For example, Qwen with the checklist achieves within-one-point agreement of 0.96 on synthetic profiles versus 0.86 on historical profiles, with Spearman correlations of 0.81 and 0.47, respectively. However, the pattern is not universal: Phi-4 with facts has equal within-one-point agreement on both profile types (0.83), while Phi-4 with the checklist has a higher Spearman correlation on historical profiles (0.52 versus 0.39). Human yes/no kappa is higher on historical profiles (0.23 against 0.11), which may reflect the larger share of inferred attributes, where invented personalisation and overstated certainty are easier to spot.

TABLE XII
RQ2: AGREEMENT BY PROFILES (H = HISTORICAL, S = SYNTHETIC).

Rater	Yes/no kappa		Within 1		Spearman	
	H	S	H	S	H	S
Human vs human	0.23	0.11	0.83	0.91	0.24	0.44
Gemma, facts	0.00	0.37	0.77	0.90	0.31	0.51
Gemma, checklist	0.06	0.37	0.73	0.91	0.28	0.48
Gemma, no facts	-0.03	0.37	0.65	0.81	-0.18	0.25
Phi-4, facts	0.32	0.24	0.83	0.83	0.49	0.52
Phi-4, checklist	0.13	0.41	0.80	0.85	0.52	0.39
Phi-4, no facts	0.10	0.29	0.77	0.58	0.45	0.04
Qwen, facts	0.19	0.18	0.83	0.81	0.47	0.52
Qwen, checklist	0.21	0.37	0.86	0.96	0.47	0.81
Qwen, no facts	0.13	0.40	0.84	0.53	0.33	0.20

E. RQ3: ranking of explanation techniques

Table XIII gives the mean overall score of each technique, with two items per technique and rater. All three humans rate direct LLM explanations worst (1.5–2.0) and the grounded technique with self-check best or tied best (3.5–4.5). Human problem rates agree: 0.67 of the violation questions are flagged for direct LLM texts on average, against 0.21 for grounded,

0.17 for grounded with check and 0.00 for templates. Templates are free of errors but are not always rated highest, because the annotators found them stiff and less personal, and one annotator gave them only 2.0.

TABLE XIII
RQ3: MEAN OVERALL SCORE (1–5) AND SPEARMAN CORRELATION

Rater	Templ.	Direct	Grounded	G.+check	ρ
Human A1	2.0	2.0	3.0	3.5	–
Human A2	4.0	2.0	3.5	4.5	–
Human A3	3.5	1.5	3.5	4.5	–
Human mean	3.2	1.8	3.3	4.2	–
Gemma, facts	3.5	3.0	3.5	3.5	0.77
Gemma, checklist	3.5	2.5	3.0	4.5	0.80
Gemma, no facts	3.5	3.5	3.0	3.0	-0.89
Phi-4, facts	4.0	2.5	3.5	3.5	0.32
Phi-4, checklist	4.0	2.0	3.5	3.5	0.32
Phi-4, no facts	4.0	2.5	3.0	2.5	-0.11
Qwen, facts	3.0	2.0	3.0	4.5	0.95
Qwen, checklist	4.5	1.5	3.5	5.0	0.80
Qwen, no facts	3.5	2.5	2.5	2.5	-0.26

The judges show mixed agreement with the human ranking. Gemma with facts rates direct explanations 3.0, while its checklist variant gives them 2.5; without facts, Gemma rates direct explanations 3.5 and grounded explanations with self-check 3.0. Phi-4 with facts and its checklist variant both rate the grounded technique and the grounded technique with self-check equally (3.5), while Phi-4 with no facts rates the latter lower (2.5). Qwen with facts reproduces the human ranking most closely (Spearman $\rho = 0.95$), whereas its checklist variant has a correlation of 0.80 and its no-facts variant has a negative correlation of -0.26. Gemma with no facts also has a negative correlation (-0.89), and Phi-4 with no facts has a correlation of -0.11. These results suggest that removing the facts block can substantially alter technique rankings, although the direction and magnitude vary across models.

Two caveats apply: with four techniques and two items each, a single item changes a mean by 0.5 to 1.0 points, and Qwen audited technique 4, so its high rating of that technique may be partly self-agreement. Human A2 and A3 agree with Qwen on technique 4 (4.5), so there is no clear sign of bias, but this needs more items to test. We therefore recommend reporting RQ3 with and without Qwen in the main round.

F. RQ4: detection of injected errors

Only two of the 15 diagnostic cases were rated in the pilot, both of type invented preference (Fig. 13). Humans flagged the injected invented preference in 83% of ratings (individual annotators 0.5–1.0), and all judge setups flagged it in at least half of the cases (Gemma with facts: 0.5; all others: 1.0). The comparison is not informative yet: the other three error types (wrong nutrition, ignored time limit, ignored dietary constraint) were not rated by anyone since they were missing in the pilot annotation form, and the normal cases already have a "yes" rate of 0.13–0.38 for invented personalization among judges, so a high detection rate may also reflect a high base rate of flags. A fair test needs the 15 diagnostics rated by humans and the false-positive rate on matched normal cases.

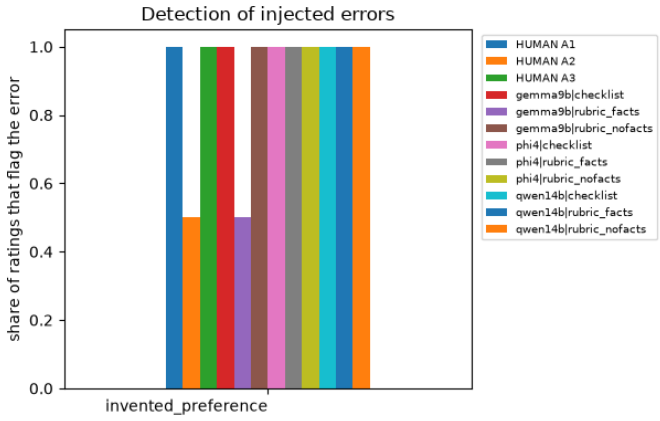


Fig. 13. Share of ratings that flag the injected error, by rater and error type (pilot: invented preference only).

G. Examples: where judges and humans agree and disagree

Numbers such as kappa or "within one point" summarise many ratings at once. To show what they mean in practice, we look at individual explanations in full (see Table XIV): one where the LLM judges came closest to the humans, one where they were furthest, and one where the humans disagreed among themselves. All examples come from the 10-item pilot, so each explanation was rated by three human annotators (A1–A3) and by nine judge setups (three judges: Gemma, Phi-4 and Qwen, each in the *checklist*, *facts* and *no-facts* setup). The examples illustrate the analysis; with one case per situation they are not statistical evidence.

How the cases were chosen and how to read the tables. For every case we take the average of the three human scores as the "human view". For each judge setup we count which of the five 1–5 scores (clarity, profile fit, goal alignment, trade-off, overall) lie within one point of that human view, and average this over the nine setups (*closeness*, 1.0 means all scores are close). The closest case is `C-A65BE` (closeness 0.92) and the furthest is `C-16004` (0.64). `C-C9CF9` is the case where the humans disagree most (overall scores from 1 to 5). A fourth case, `C-A876B`, is used only for comparison. In the tables, scores are on a 1–5 scale (5 is best); a "Y" means that the rater found the problem (for example an invented preference) and "n" that no problem was found.
Human agree. is the share of human score pairs that differ by at most one point.

1) Judges closest to humans: `C-A65BE` (grounded + self-check, historical profile):

> *Based on your past reviews, you tend to like recipes that have a lot of sodium, fiber, and protein. However, the recipe "Potato Nachos" does not feature any of the ingredients you've shown to like in the past. It also has lower sodium levels than your usual preference. On the other hand, it does have a high*

TABLE XIV

THE SELECTED CASES. "OVERALL" IS THE 1–5 OVERALL SCORE. CLOSENESS: SHARE OF THE JUDGES' SCORES WITHIN ONE POINT OF THE HUMAN AVERAGE, AVERAGED OVER NINE SETUPS. FLAG MATCH: SHARE OF THE FIVE PROBLEM QUESTIONS ON WHICH A JUDGE GAVE THE SAME ANSWER AS THE HUMAN MAJORITY.

Case	Technique (profile)	Human overall	Judge overall	Human agree.	Close-ness	Flag match
C-A65BE	grounded + check (hist.)	3, 4, 4	3–5	0.75	0.92	0.76
C-16004	direct LLM (hist.)	2, 2, 1	1–3	0.92	0.64	0.56
C-C9CF9	template (hist.)	1, 5, 4	2–5	0.50	0.81	0.78
C-A876B	template (synth.)	3, 3, 3	3–5	0.83	0.72	0.89

TABLE XV

OVERALL SCORE (1–5) GIVEN TO EACH EXPLANATION BY THE THREE HUMANS AND BY EACH JUDGE SETUP (CL = CHECKLIST, F = RUBRIC WITH FACTS, NF = RUBRIC WITHOUT FACTS). THE CLOSER A JUDGE IS TO THE HUMAN SCORES, THE BETTER IT MATCHES.

	Humans			Gemma 2 9B			Phi-4			Qwen 2.5 14B		
Case	A1	A2	A3	CL	F	NF	CL	F	NF	CL	F	NF
C-A65BE (grounded + check)	3	4	4	4	3	3	4	4	3	5	4	3
C-16004 (direct LLM)	2	2	1	3	3	3	2	2	3	1	1	3
C-C9CF9 (template)	1	5	4	3	3	3	4	4	3	5	3	2
C-A876B (template, synthetic)	3	3	3	4	4	4	4	4	5	4	3	5

amount of fiber, which aligns with your past pattern. The recipe takes about 55 minutes to prepare, which is slightly longer than your typical 40 minutes.

What the humans said. The three annotators gave overall scores of 3, 4 and 4. None marked a factual error and the majority saw no invented preference or unsupported claim. Two of three marked *overstated certainty*: the text says "you tend to like…" about a person whose tastes were only *inferred* from old reviews. The humans disagreed on how well the text fits the person (profile fit 2, 4 and 5).

What the judges said. Eight of the nine judge setups gave an overall score of 3 or 4; only Qwen with the checklist gave 5. Judges and humans also gave the same answers on the main problem questions (no factual error, no invented preference).

Why they agree, in plain words. The explanation is short, names its own weaknesses (it admits that the recipe does not contain the user's usual ingredients and is longer than usual) and uses only facts that were checked. When an explanation is honest about downsides, humans and judges alike recognize it as decent but not perfect. **Where they still differ:** the subtle problem, claiming certainty about an inferred taste, was caught by only two of the nine judge setups (Qwen with and without facts). The "agreement" on this case is therefore agreement that *nothing obvious is wrong*, not that the judges see everything the humans see.

2) Judges furthest from humans: C-16004 (direct LLM, historical profile):

I think you'll enjoy this recipe. This is Alice Springs Chicken, it features a combination of flavors and ingredients that are commonly liked by you, such as cheese and a savory sauce. The recipe also includes chicken, which is a staple protein in many of the dishes you've liked in the past. While it does take a bit longer to prepare than some of your other favorites, it is likely to be worth the extra time.

TABLE XVI

PROBLEM FLAGS FOR C-16004. Y = PROBLEM FOUND, N = NOT FOUND. CL = CHECKLIST, F = WITH FACTS, NF = WITHOUT FACTS.

	Humans			Gemma			Phi-4			Qwen		
Flag	A1	A2	A3	CL	F	NF	CL	F	NF	CL	F	NF
Factual error	Y	Y	Y	n	n	n	n	Y	n	n	Y	n
Invented preference	Y	Y	Y	n	n	n	Y	Y	n	Y	Y	Y
Unsupported claim	Y	Y	Y	Y	n	n	Y	Y	Y	Y	Y	Y

What the humans said. All three annotators agreed that this is a poor explanation: overall 2, 2 and 1, and all three marked a factual error, an invented personalization and an unsupported claim. They also rated its honesty about downsides as very low (trade-off 1, 2 and 1).

What the judges said. Table XVI shows the problem flags. Gemma gave an overall score of 3 in all three setups and found neither the factual error nor the invented preference in any of them. Phi-4 and Qwen gave scores close to the humans (1–2) when they were given the checked recipe facts or used the checklist, but *without the facts* both rated the text 3, a full point or more above the humans. The factual error was caught only by the two judge setups that saw the checked facts (Phi-4 and Qwen with facts), 2 of 9; the invented preference by 5 of 9; the unsupported claim by 7 of 9.

Why they disagree, in plain words. This text is fluent and confident, so it looks good. Whether its claims are *true* can only be seen by comparing them with the recipe's real data. A judge without that data, or one that does not check claims one by one, is easily convinced by confident writing. This single case reproduces our main result (Section IV): judges need the checked facts, and unchecked LLM text (the direct technique) is the kind of text most likely to hide errors from a careless reader, human or machine.

3) Humans disagree among themselves: C-C9CF9 (template, historical profile):

Creamy Oven-Baked Risotto is one option we can offer you. It does not clearly match the ingredients and themes you usually favour. It takes about 40 minutes. It differs from the usual pattern of your favourites, which tend to have longer cooking times. Per serving it has about 282 kcal, 10 g of protein and 584 mg of sodium.

What the humans said. The three annotators agreed on all problem questions (no factual error, no invented preference, no unsupported claim), so the text is accurate. But their overall scores were 1, 5 and 4, and their profile-fit scores 1, 5 and 5. One annotator penalized the explanation for recommending something that, by its own admission, does not match the person; the other two rewarded it for being accurate and honest.

What the judges said. Overall scores ranged from 2 to 5 (Gemma 3, 3, 3; Phi-4 4, 4, 3; Qwen 5, 3, 2). They spread over the same range as the humans, and the average of the judges (about 3.3) is almost exactly the human average (3.33). Individual judge setups, however, can sit far from it (Qwen with the checklist gave 5; Qwen without facts gave 2).

Why this matters, in plain words. When even the humans do not agree on what a good score is, there is no single "right answer" for a judge to match. The question here is partly a matter of taste: is a truthful but cold explanation of a poor match good or bad? This is why the agreement between humans (Table X) is the ceiling for judges, and why we compare judges with the human-versus-human baseline and not with perfection.

4) Which techniques did judges follow, and which did they not?: Table XV suggests a pattern across the four explanation styles, with the caveat that each rests on a single case.

- **Grounded + self-check (`C-A65BE`): agreement.** Judges and humans gave similar scores and similar flags. The text is accurate, honest about its trade-offs and easy to verify.
- **Direct LLM (`C-16004`): agreement only when the judge sees the facts.** With the checked facts (and for Phi-4 and Qwen also with the checklist) the judges recognised the errors; without them, or for Gemma, they were too generous.
- **Template (`C-C9CF9`, `C-A876B`): agreement on facts, disagreement on quality.** Template text is accurate, so judges and humans agree that there is no factual problem. Scores are another matter. In `C-A876B` the template lists 827 kcal and 3,709 mg of sodium without saying that these are high; two of three humans gave the trade-off question a 1 (average 1.7), whereas the judges gave it 3 to 5 and rated the explanation 3 to 5 overall (humans: 3, 3, 3). The judges therefore *rewarded the presence of numbers* and missed the *absence of an honest warning*.

These observations are hypotheses for the main annotation round, which will contain 120 explanations per technique and make a technique-level comparison possible (RQ3).

Take-aways for a non-specialist reader. (i) An LLM judge is a fast but imperfect reader: it usually notices bad writing and obvious invented claims, but is easily impressed by confident text. (ii) Showing it the checked facts makes it more careful, and in the example this made the difference between catching and missing a factual error. (iii) Where humans themselves disagree, a judge cannot be expected to be "right". (iv) Judges tend to be more lenient than humans about missing warnings (trade-offs). For these reasons we propose using LLM judges to *screen* explanations and to flag the suspicious ones for human review, not to replace the human check entirely.

V. Overall Discussion

Findings in context. The pilot shows that rubric-based LLM judges with access to checked facts can reach human-level agreement on scale questions and approach it on yes/no questions, which is consistent with the general evidence that judges correlate with human judgments [6], [8]. The strong dependence of the result on the judge's calibration (Gemma almost never flags problems, Qwen flags many) agrees with the view that agreement statistics alone hide systematic bias, and that error-type-specific measures such as yes-rate, precision and recall are necessary. The weak human agreement on yes/no questions (kappa 0.22) shows that the reference is itself uncertain, which is the main reason to collect multiple annotators and to report intervals.

Limitations. (i) Only 10 of 510 items were rated by humans, so all comparisons have wide intervals and rating pairs are not independent; (ii) human yes/no questions have low prevalence (ignored constraint 3%), which makes kappa unstable; (iii) all judge and generator models are small open-weight models run locally, and Qwen both audited and judged; (iv) the first-person phrasing of direct LLM texts may unblind technique; (v) profiles inferred from history are not confirmed by real users; (vi) technique 4 equals technique 3 in 69% of cases; (vii) the Notebook 9 results were produced in one run with one repetition, so judge variability across repeats is not yet measured; (viii) confidence intervals resample items, whereas the plan asks for profile-level resampling because items of one profile are related.

Plan coverage. Table XVII compares what was specified with what was done.

Further work. The next step is the main annotation round, with one annotator on a stratified subset first and three annotators later, after a calibration session on the most disputed pilot items. Then the judges would be run on all rated items with the final configuration frozen on the pilot; the three pairwise and diagnostic parts need human labels to test RQ4 and position bias. We consider the following aspects novel and worth pursuing: the pairing of inferred and fully specified profiles, the injected-error diagnostics with a hidden key, the ablation of the facts block, and the self-check generation technique.

VI. Conclusion

We built FoodJudgeBench, an end-to-end and reproducible benchmark for comparing LLM judges with human annotators on personalized food recommendation explanations. The pipeline covers data audit, 60 profiles, a recommendation

TABLE XVII
COVERAGE OF THE PROJECT SPECIFICATION.

Plan step	Status	What was done and what is missing
1. Research objectives	Done	Four research questions defined; scope limited to assessment reliability.
2. Data Audit and catalog	Done	Full audit; authorship rows separated; catalog with percentiles and diet labels; missing values not imputed. Only a modification date exists, so a temporal hold-out is limited
3. Profile collections	Done	30 historical and 30 synthetic profiles with status and evidence.
4. Recommendation pool	Done	Content-based pool with weak and violating candidates and recorded signals.
5. Explanation techniques	Done	Four of six techniques (template, direct LLM, grounded LLM, grounded with self-check) done as per the research plan. Content-based, model-based and contrastive explanations were not implemented; neutral and goal-framed wording exists for 17 cases, but only neutral text was annotated.
6. Evaluation cases	Done	480 blinded cases, 15 diagnostics (four error types), 15 pairs, hidden key. The unsupported social proof diagnostic of the plan was not created, and formatting artifacts were repaired.
7. Human rubric	Done	Ten questions with not-applicable handling and anchors with revised annotator guide.
8. Human annotation	Partly	Pilot with three annotators on 10 items. Main round (510 items, three annotators) not done and is out of scope for this coursework; no pairwise human labels; no nutrition-qualified annotators.
9. LLM judges	Done	Three families, three prompt setups, order reversal implemented and cached.
10. Agreement and generalization	Partly	Agreement metrics with bootstrap intervals, by source and technique. No profile-level resampling, no sparse versus rich or unseen-recipe analysis, no Recall@K or NDCG@K.
11. Release	Done (v0.1)	Case bank, answer key, rubric, ratings, results and code; recipe text withheld; dataset not redistributed.

pool, four explanation techniques, a blinded case bank of 510 items, a ten-question rubric, annotation forms, three judge families and an analysis and release notebook. On the 10-item pilot, the best judge setups agreed with humans about as well as humans agree with each other (within-one-point agreement 0.93 against 0.88, yes/no kappa 0.35 against 0.22) and reproduced the human preference for grounded explanations over direct LLM text; but intervals are wide and the evidence is preliminary. The hardest parts of the work were the data issues in HUMMUS (authorship rows, placeholder ratings, sparse histories), keeping the evaluation blind, and fine-tuning the small local models to return valid structured judgments. We gained practical experience with Exploratory Data Analysis, LLM generation and evaluation, annotations, agreement statistics and reproducible pipelines. The main way forward is to collect the full human annotations and run the judges under a frozen configuration on all items.

ACKNOWLEDGMENT

We thank our project supervisor, Dr Mehrdad Rostami, for the feedback and guidance on scoping the annotation work.

USE OF AI

We used Claude to understand the project plan and debug and refine our code. The project setup, all the Jupyter notebooks, methodological choices, data annotations, and final interpretations of the results are our own work and reasoning. We finalised the methodology after detailed discussions with the project supervisor. We also provided our initial results and analysis to Claude to generate the first draft of this report. Then we independently restructured, revised, and expanded the report over multiple iterations, adding new results and deeper interpretations. Also, Grammarly was used for paraphrasing and grammatical corrections of this report.

REFERENCES

[1] W. Min, S. Jiang, and R. Jain, "Food recommendation: Framework, existing solutions, and challenges," IEEE Trans. Multimedia, vol. 22, no. 10, pp. 2659–2671, 2020.
[2] C. Trattner and D. Elsweiler, "Investigating the healthiness of internet-sourced recipes: Implications for meal planning and recommender systems," in Proc. 26th Int. Conf. World Wide Web (WWW), 2017, pp. 489–498.
[3] N. Tintarev and J. Masthoff, "Explaining recommendations: Design and evaluation," in Recommender Systems Handbook, 2nd ed., F. Ricci, L. Rokach, and B. Shapira, Eds. New York: Springer, 2015, pp. 353–382.
[4] Y. Zhang and X. Chen, "Explainable recommendation: A survey and new perspectives," Found. Trends Inf. Retr., vol. 14, no. 1, pp. 1–101, 2020.
[5] R. H. Thaler and C. R. Sunstein, Nudge: Improving Decisions About Health, Wealth, and Happiness. New Haven, CT: Yale Univ. Press, 2008.
[6] L. Zheng et al., "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets and Benchmarks Track, 2023.
[7] J. Gu et al., "A survey on LLM-as-a-judge," arXiv:2411.15594, 2024.
[8] Y. Liu et al., "G-Eval: NLG evaluation using GPT-4 with better human alignment," in Proc. Conf. Empirical Methods Natural Language Process. (EMNLP), 2023, pp. 2511–2522.
[9] S. Kim et al., "Prometheus: Inducing fine-grained evaluation capability in language models," in Proc. Int. Conf. Learning Representations (ICLR), 2024.
[10] P. Wang et al., "Large language models are not fair evaluators," arXiv:2305.17926, 2023.
[11] A. Panickssery, S. R. Bowman, and S. Feng, "LLM evaluators recognize and favor their own generations," arXiv:2404.13076, 2024.
[12] F. Bölz et al., "HUMMUS: A linked, healthiness-aware, user-centered and argument-enabling recipe data set for recommendation," in Proc. 17th ACM Conf. Recommender Systems (RecSys), 2023. Data: https://gitlab.com/felix134/connected-recipe-data-set.
[13] B. P. Majumder, S. Li, J. Ni, and J. McAuley, "Generating personalized recipes from historical user preferences," in Proc. Conf. Empirical Methods Natural Language Process. and Int. Joint Conf. Natural Language Process. (EMNLP-IJCNLP), 2019, pp. 5976–5982.
[14] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Inf. Process. Manage., vol. 24, no. 5, pp. 513–523, 1988.
[15] A. Dubey et al., "The Llama 3 herd of models," arXiv:2407.21783, 2024.
[16] A. Yang et al., "Qwen2.5 technical report," arXiv:2412.15115, 2024.
[17] K. Krippendorff, Content Analysis: An Introduction to Its Methodology, 4th ed. Thousand Oaks, CA: SAGE, 2018.
[18] J. Cohen, "A coefficient of agreement for nominal scales," Educ. Psychol. Meas., vol. 20, no. 1, pp. 37–46, 1960.
[19] Gemma Team, "Gemma 2: Improving open language models at a practical size," arXiv:2408.00118, 2024.
[20] M. Abdin et al., "Phi-4 technical report," arXiv:2412.08905, 2024.
[21] B. Efron, "Bootstrap methods: Another look at the jackknife," Ann. Statist., vol. 7, no. 1, pp. 1–26, 1979.

APPENDIX A
RUBRIC

TABLE XVIII
RUBRIC QUESTIONS (NOTEBOOK 6, `RUBRIC.JSON`).

Question	Answers	Meaning
Ignored constraint	yes/no/na	Contradicts a stated diet, allergy or time limit
Factual error	yes/no	Wrong statement about the recipe
Invented personalization	yes/no	Claims something about the person that the profile does not say
Unsupported claim	yes/no	States as fact what cannot be checked
Overstated certainty	yes/no/na	Presents an inferred attribute as certain
Clarity	1–5	Easy to read and understand
Profile fit	1–5	Uses the relevant profile information correctly
Goal alignment	1–5/na	Addresses the stated health goal, including honest mismatch
Trade-off	1–5/na	Honestly communicates downsides
Overall	1–5	Overall suitability of the explanation

APPENDIX B
ANNOTATIONS WEBFORM

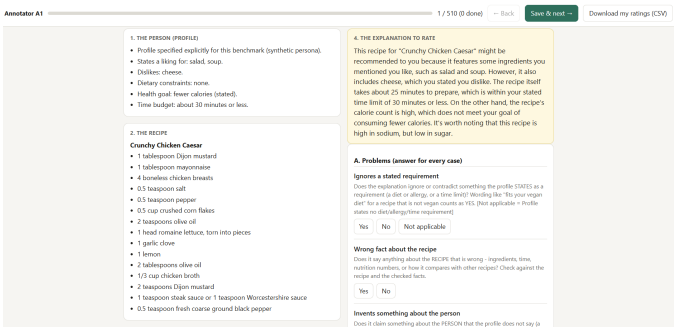


Fig. 14. Annotator webform

APPENDIX C
JUDGE CONFIGURATION

TABLE XIX
JUDGE CONFIGURATION OF THE REPORTED RUN.

Setting	Value
Judge models	gemma2:9b, qwen2.5:14b, phi4 (Ollama)
Generator / auditor	llama3.1:8b / qwen2.5:14b
Temperature, seed	0.1 (judges), 0.4 (generation); 42
Context length	4,096 tokens
Initial Setups run	rubric_facts, checklist
Final Setup run	rubric_nofacts, rubric_facts, checklist
Repetitions	1
Items judged	10 (8 normal, 2 diagnostics)
Parse failures / repairs	0 / 0

APPENDIX D
EXAMPLE DIAGNOSTIC CASES

Examples of injected errors (Notebook 6): *invented preference* ("You mentioned that you enjoy long weekend cooking projects, which makes this a good fit"), *ignored time limit* (a 200-minute recipe described as fitting a 45-minute limit) and *ignored dietary constraint* (a recipe violating a vegan requirement described as fitting it).

APPENDIX E
REPRODUCIBILITY

All commands, library versions and the mapping of tables to notebooks are in the repository README. Tables XI–XIII are produced by `10_Analysis_and_Release.ipynb` (files `rq1_overall.csv`, `rq2_by_source.csv`, `rq3_technique_means.csv`, `rq4_detection.csv`); Table X by `8_Annotation_Collection.ipynb`; Table IX by `5_Building_Explanations.ipynb`; Table I by `1_HUMMUS_EDA.ipynb`.